AP-E3: A Microscopic Origin Candidate for Projective Level Magnitude Two
Two-Orbital Mott–Hund Locking, the Quadratic Veronese Line, and the Exactly-Two UV Obstruction

Route F research note

14 July 2026

Abstract

This note asks which microscopic statement can produce the level magnitude $|k| = 2$ needed for a three-state projective carrier, rather than merely inserting that integer in an effective action. Large $U(1)$ gauge invariance proves only $k \in \mathbb{Z}$ and cannot distinguish 2 from 1 or 3. We then construct a minimal conditional candidate. Two spinful bosonic orbitals are placed on a unit-occupancy Mott plateau and coupled ferromagnetically. Their low-energy Hilbert space is the symmetric spin-one triplet. For a normalized doublet s, its oriented ground state is

$$|s; 2\rangle = |s\rangle \otimes |s\rangle, \qquad -\mathrm{i}\langle s; 2|\mathrm{d}(s; 2)\rangle = 2\big(-\mathrm{i}s^\dagger \mathrm{d}s\big).$$

In standard algebraic-geometric conventions the ket eigenline is $L_{\text{ket}} = \nu_2^* \mathcal{O}_{\mathbb{CP}^2}(-1) = \mathcal{O}_{\mathbb{CP}^1}(-2)$, whereas its dual prequantum line is $L_{\text{pre}} = L_{\text{ket}}^* = \mathcal{O}_{\mathbb{CP}^1}(2)$. Thus the declared microscopic dimer derives $|k| = 2$ and $H^0(\mathbb{CP}^1, L_{\text{pre}}) \simeq \mathrm{Sym}^2\, \mathbb{C}^2$ with dimension three, without a bare level-two counterterm. With the convention $A = -\mathrm{i}\langle s|\mathrm{d}s\rangle$, however, the displayed microscopic kinetic term gives signed level $k = -2$; reversing the orientation gives $k = +2$.

The construction is verified by a deterministic 27/27 audit. At $U = 4, \mu = 1.5, J_H = 1, h = 0.2$, the sufficient interacting-plateau inequality has margin 2.025, and the complete spinful occupancy problem has the unique sector $(N_1, N_2) = (1, 1)$, charge gap 1.85, Hund spectrum $(-1/4, -1/4, -1/4, 3/4)$, and a numerical Chern sequence converging to 2.000000501994128. The result is nevertheless conditional: the repository does not yet derive why the UV field content contains exactly two same-orientation constituents, why singleton or odd sectors are forbidden, or why the resulting triplet is chiral Route-E family data. The physical orientation selecting the signed Route-E level is also not yet derived. Consequently AP-E3 has a complete candidate-level magnitude derivation but not a complete signed UV physics derivation, and `physics_promotion_allowed=false`.

Contents

1 Question, assumptions, and precise status

AP-E1 derived the Hopf quotient and showed that a single projective doublet naturally gives a ket line $\mathcal{O}(-1)$ and dual prequantum line $\mathcal{O}(1)$, not their tensor squares. AP-E2 froze the independent discriminant–Killing theorem but did not produce a Berry level. AP-E3 must therefore answer a logically separate question: what microscopic Hilbert space, Hamiltonian, and gap produce the tensor-square Berry line, and what additional principle makes that sector the lowest physical nonzero one?

The selected candidate uses the following declared data.

(A1) There are two orbitals $r = 1, 2$, each with a spinor boson a_{rA}, $A = 1, 2$.

(A2) The full interacting core Hamiltonian lies on the unit-occupancy plateau. A sufficient condition derived below is $C := \mu + h/2 + J_H/4 < U - J_H/8$.

(A3) A ferromagnetic Hund interaction locks the two spin-1/2 moments into their symmetric triplet.

(A4) A weak orientation field selects a unique coherent ground-state line; all portal, drive, and thermal scales remain below the retained gaps.

These are microscopic model inputs, not consequences of Route E.

Theorem 1.1 (Conditional AP-E3 level-two-magnitude theorem)**.** *Under assumptions (A1)–(A4), and while the many-body gap remains open, the ground-state ket line and its dual prequantum line obey*

$$L_{\text{ket}} \simeq \mathcal{O}_{\mathbb{CP}^1}(-2), \qquad L_{\text{pre}} = L_{\text{ket}}^* \simeq \mathcal{O}_{\mathbb{CP}^1}(2),$$

with $c_1(L_{\mathrm{ket}}) = -2$*,* $c_1(L_{\mathrm{pre}}) = 2$*, and*

$$H^0(\mathbb{CP}^1, L_{\mathrm{pre}}) \simeq \mathrm{Sym}^2\,\mathbb{C}^2, \qquad \dim = 3.$$

Relative to $S_k = \hbar k \int A$*, the aligned Hamiltonian written below has* $k = -2$*; the oppositely oriented Hamiltonian has* $k = +2$*. Hence the model derives* $|k| = 2$*, not the physical sign, without a bare level-two term.*

Fail-closed gate 1.2 (What the theorem does not prove)**.** The theorem does not derive (A1), does not impose a fundamental superselection against all single-constituent excitations, and does not identify a spin-one internal multiplet with three chiral families. A Mott gap makes singleton sectors irrelevant to the strict low-energy EFT; it does not erase them from the full UV Hilbert space. A fundamental claim that magnitude two is the lowest nonzero level therefore still requires an exactly-two UV rule or an independent odd-sector prohibition. It also does not derive which orientation is the chiral Route-E convention, so the signed level remains a gate.

2 Large gauge invariance is an integrality theorem, not a selector

Let $s \in \mathbb{C}^2$ satisfy $s^\dagger s = 1$, with redundancy $s \sim e^{\mathrm{i}\alpha} s$. The Hopf connection and curvature are

$$A = -\mathrm{i} s^\dagger \mathrm{d}s, \qquad F = \mathrm{d}A, \qquad \frac{1}{2\pi}\int_{\mathbb{CP}^1} F = 1. \tag{1}$$

For a closed worldline γ, consider

$$S_k = \hbar k \oint_\gamma A. \tag{2}$$

Under $s \mapsto e^{\mathrm{i}\alpha} s$, one has $A \mapsto A + \mathrm{d}\alpha$, so a large transformation of winding m changes the action by

$$\Delta S_k = \hbar k \oint_\gamma \mathrm{d}\alpha = 2\pi \hbar k m. \tag{3}$$

Single-valuedness of $e^{\mathrm{i}S/\hbar}$ gives exactly

$$\boxed{k \in \mathbb{Z}.} \tag{4}$$

Levels $1, 2, 3$ all pass this condition. No argument based only on large gauge invariance can select $k = 2$.

There are two common traps.

1. The transformation $s \mapsto -s$ is already the gauge transformation $\alpha = \pi$; by itself it does not require even k.
2. Declaring a field to have gauge weight two need not double the faithful Hopf level. If the Gauss law is $2N = K$, then the physical occupation is $N = K/2$; the minimal label $K = 2$ can still describe one Hopf quantum. A genuine $\mathcal{O}(2)$ line requires a tensor square, two additive occupied eigenlines, or another independently normalized topological term.

An independent exact even-parity projector

$$P_{\mathrm{even}} = \frac{1 + (-1)^N}{2} \tag{5}$$

would remove odd occupations. For $E_N = UN(N-1)/2 - \mu N$, the lowest nonzero even sector is uniquely $N = 2$ when

$$\frac{U}{2} < \mu < \frac{5U}{2}. \tag{6}$$

But the parity projector and the plateau are new microscopic inputs; Eq. (4) does not generate them.

3 The two-orbital Mott–Hund model

3.1 Microscopic action

Let the number and spin operators be

$$N_r = a_{rA}^\dagger a_{rA}, \qquad \boldsymbol{S}_r = \frac{1}{2} a_{rA}^\dagger \boldsymbol{\sigma}_{AB} a_{rB}. \tag{7}$$

Repeated spinor indices are summed. The proposed microscopic action is

$$S_{\text{micro}} = \int \mathrm{d}t \left[\mathrm{i}\hbar \sum_{r,A} a_{rA}^\dagger \dot{a}_{rA} - H_{\text{micro}}(\boldsymbol{n}) \right], \tag{8}$$

$$H_{\text{micro}}(\boldsymbol{n}) = \sum_{r=1}^{2} \left[\frac{U}{2} N_r(N_r - 1) - \mu N_r \right] - J_H \boldsymbol{S}_1 \cdot \boldsymbol{S}_2 - h\boldsymbol{n} \cdot (\boldsymbol{S}_1 + \boldsymbol{S}_2) + H_{\text{portal}}. \tag{9}$$

Here $U, J_H, h > 0$, and $\boldsymbol{n} \in S^2$ is an adiabatic orientation parameter. The two-orbital structure is explicit rather than hidden in a level coefficient. Similar Mott/Hund and coherent-state techniques are standard, but the specific exactly-two proposal here is a repository model, not a theorem imported from the recent literature [1, 2, 3].

3.2 Bare unit-occupancy plateau

For one orbital the onsite energy is

$$E_n^{(0)} = \frac{U}{2} n(n-1) - \mu n, \qquad n \in \mathbb{Z}_{\geq 0}. \tag{10}$$

If $0 < \mu < U$, then $E_0^{(0)} - E_1^{(0)} = \mu > 0$, and, for $n \geq 2$,

$$E_n^{(0)} - E_1^{(0)} = (n-1) \left(\frac{Un}{2} - \mu \right) \geq (n-1)(U - \mu) > 0. \tag{11}$$

Thus $n = 1$ is the unique onsite ground occupation and the bare charge gap is

$$\Delta_{\text{ch}}^{(0)} = \min\{\mu, U - \mu\}. \tag{12}$$

3.3 Exact interacting occupancy minimum and coercive tail

Set $H_{\text{portal}} = 0$ first. For a single spinful bosonic mode with occupation n_r, Bose symmetry gives $j_r = n_r/2$. For $J_H, h > 0$, the lowest state in the fixed (n_1, n_2) sector has maximal total spin aligned with $\boldsymbol{n}$. The exact core-Hamiltonian minimum is

$$E_{\min}(n_1, n_2) = \sum_{r=1}^{2} \left[\frac{U}{2} n_r(n_r - 1) - \mu n_r \right] - \frac{J_H}{4} n_1 n_2 - \frac{h}{2}(n_1 + n_2). \tag{13}$$

The bare window $0 < \mu < U$ is not by itself an interacting theorem. For example, $(U, \mu, J_H, h) = (1, 0.99, 3.9, 0)$ satisfies both $0 < \mu < U$ and $4U > J_H$, but Eq. (13) gives

$$E_{\min}(1, 1) = -2.955, \qquad E_{\min}(2, 2) = -5.86.$$

The following inequality repairs that logical gap.

Proposition 3.1 (Sufficient global interacting plateau)**.** *Assume $U, \mu, J_H, h > 0$, set*

$$C := \mu + \frac{h}{2} + \frac{J_H}{4}, \tag{14}$$

and suppose

$$\boxed{C < U - \frac{J_H}{8}} \qquad \Longleftrightarrow \qquad \boxed{0 < \mu < U - \frac{h}{2} - \frac{3J_H}{8}}. \tag{15}$$

Then $(n_1, n_2) = (1, 1)$ is the unique global minimizer of Eq. (13) over $\mathbb{Z}^2_{\geq 0}$. This condition is sufficient, not claimed necessary.

Proof. Put $x = n_1 - 1$ and $y = n_2 - 1$, so $x, y \in \mathbb{Z}_{\geq -1}$. Direct subtraction gives

$$\begin{aligned} D(x, y) &:= E_{\min}(1 + x, 1 + y) - E_{\min}(1, 1) \\ &= \frac{U}{2}\{x(x + 1) + y(y + 1)\} - C(x + y) - \frac{J_H}{4}xy. \end{aligned} \tag{16}$$

First take $x, y \geq 0$. At fixed $s = x + y$, Eq. (16) decreases as xy increases, so its minimum occurs at the most balanced integer pair. For $s = 2m$, $m \geq 1$, that minimum is

$$D(m, m) = m\left[\left(U - \frac{J_H}{4}\right)m + U - 2C\right] > 0, \tag{17}$$

because the bracket is increasing in m and at $m = 1$ equals $2(U - J_H/8 - C) > 0$. Condition (15) also implies $J_H < 8U/3 < 4U$, hence $U - J_H/4 > 0$. For $s = 2m + 1$, the balanced value is

$$D(m, m + 1) = \left(U - \frac{J_H}{4}\right)(m^2 + m) + \frac{U}{2} + \left(\frac{U}{2} - C\right)(2m + 1). \tag{18}$$

It starts at $U - C > 0$, and its successive difference is $2(m + 1)(U - J_H/4) + U - 2C > 0$; hence it too is positive.

It remains to check the boundary allowed by $x, y \geq -1$. For $y \geq 0$,

$$D(-1, y) = \frac{U}{2}y^2 + \left(\frac{U}{2} - C + \frac{J_H}{4}\right)y + C. \tag{19}$$

At $y = 0$ this is $C > 0$; at $y = 1$ it is $U + J_H/4 > 0$, and for $y \geq 1$ its successive difference $Uy + U - C + J_H/4$ is positive. The case $D(x, -1)$ is symmetric, while $D(-1, -1) = 2C - J_H/4 = 2\mu + h + J_H/4 > 0$. Every lattice point other than $(x, y) = (0, 0)$ therefore has strictly higher energy. □

Let $N = n_1 + n_2$. From $n_1^2 + n_2^2 \geq N^2/2$ and $n_1 n_2 \leq N^2/4$,

$$\boxed{E_{\min}(n_1, n_2) \geq \frac{4U - J_H}{16}N^2 - \left(\frac{U}{2} + \mu + \frac{h}{2}\right)N.} \tag{20}$$

Hence $4U > J_H$ makes the high-occupancy tail coercive. Notice that Eq. (15) already implies this inequality. A finite sector enumeration together with Eq. (20) proves the global ground sector at any stated benchmark.

At

$$(U, \mu, J_H, h) = (4, 1.5, 1, 0.2) \tag{21}$$

one has

$$C = 1.85 < 3.875 = U - \frac{J_H}{8}, \qquad (U - J_H/8) - C = 2.025,$$

and

$$E_{\min}(1,1) = -3.45, \qquad E_{\min}(1,0) = E_{\min}(0,1) = -1.60.$$

The exact scan for $N < 8$ has the unique ground $(1,1)$ and interacting charge gap 1.85. For $N \geq 8$, the lower bound is increasing and is already 31.2, excluding every omitted occupancy sector.

4 Exact triplet projection and the orientation gap

Inside $N_1 = N_2 = 1$, each orbital carries spin 1/2. With $\boldsymbol{S}_{\text{tot}} = \boldsymbol{S}_1 + \boldsymbol{S}_2$,

$$\boldsymbol{S}_1 \cdot \boldsymbol{S}_2 = \frac{1}{2}\left(\boldsymbol{S}_{\text{tot}}^2 - \frac{3}{2}\right). \tag{22}$$

The Hund term has the exact spectrum

$$E_{j=1} = -\frac{J_H}{4} \quad \text{(3-fold)}, \qquad E_{j=0} = \frac{3J_H}{4} \quad \text{(1-fold)}, \tag{23}$$

with singlet gap $\Delta_{\text{singlet}} = J_H$. The triplet projector is

$$P_{j=1} = \frac{1 + P_{12}}{2} = \frac{\boldsymbol{S}_{\text{tot}}^2}{2}, \qquad \operatorname{Tr} P_{j=1} = 3. \tag{24}$$

Thus ferromagnetic locking is precisely projection onto $\operatorname{Sym}^2 \mathbb{C}^2$.

For fixed $\boldsymbol{n}$, the orientation term gives the four energies

$$-\frac{J_H}{4} - h, \quad -\frac{J_H}{4}, \quad -\frac{J_H}{4} + h, \quad \frac{3J_H}{4}. \tag{25}$$

For $h > 0$, the ground state is unique and the triplet orientation gap is h. If $s(\boldsymbol{n})$ is the normalized $+1$ eigenvector of $\boldsymbol{n} \cdot \boldsymbol{\sigma}$, then

$$|\Omega(\boldsymbol{n})\rangle = |s(\boldsymbol{n})\rangle \otimes |s(\boldsymbol{n})\rangle. \tag{26}$$

This is the microscopic line whose topology is computed next.

5 Quadratic Veronese geometry and Berry factor two

5.1 The symmetric coherent state

Write $s = (z_0, z_1)^T$, with $|z_0|^2 + |z_1|^2 = 1$. In the normalized triplet basis,

$$|s; 2\rangle = z_0^2|1,1\rangle + \sqrt{2} z_0 z_1 |1,0\rangle + z_1^2|1,-1\rangle. \tag{27}$$

Its norm is $(|z_0|^2 + |z_1|^2)^2 = 1$, and it has no singlet component. Projectively it is the quadratic Veronese embedding

$$\nu_2 : \mathbb{CP}^1 \longrightarrow \mathbb{CP}^2, \qquad [z_0 : z_1] \longmapsto [z_0^2 : \sqrt{2} z_0 z_1 : z_1^2]. \tag{28}$$

Under $s \mapsto e^{\mathrm{i}\alpha} s$, the pair transforms as

$$|s; 2\rangle \mapsto e^{2\mathrm{i}\alpha}|s; 2\rangle. \tag{29}$$

This phase weight is a consequence of a tensor product, not a relabeling of the gauge generator.

5.2 Connection, curvature, and quantum metric

Differentiating the tensor product gives

$$\mathrm{d}|s;2\rangle = \mathrm{d}|s\rangle \otimes |s\rangle + |s\rangle \otimes \mathrm{d}|s\rangle. \tag{30}$$

Using $\langle s|s\rangle = 1$,

$$\langle s;2|\mathrm{d}(s;2)\rangle = 2\langle s|\mathrm{d}s\rangle, \tag{31}$$

and therefore

$$\boxed{A_2 = 2A_1, \qquad F_2 = 2F_1.} \tag{32}$$

The Fubini–Study quadratic form is

$$g(\mathrm{d}\psi, \mathrm{d}\psi) = \langle \mathrm{d}\psi|\mathrm{d}\psi\rangle - |\langle \psi|\mathrm{d}\psi\rangle|^2. \tag{33}$$

Substitution of the product derivative yields

$$\boxed{g_2 = 2g_1.} \tag{34}$$

Thus both symplectic and metric data are derived from the microscopic pair. This is the standard coherent-state Berry construction applied to the tensor product rather than assumed at the effective level [4, 5].

5.3 Patch proof, ket line, and dual prequantum line

Use northern and southern unit sections

$$s_N = \begin{pmatrix} \cos(\theta/2) \\ e^{\mathrm{i}\phi}\sin(\theta/2) \end{pmatrix}, \qquad A_N = \frac{1-\cos\theta}{2}\mathrm{d}\phi, \tag{35}$$

$$s_S = e^{-\mathrm{i}\phi} s_N, \qquad A_S = -\frac{1+\cos\theta}{2}\mathrm{d}\phi. \tag{36}$$

On the overlap, the pair transition is $e^{-2\mathrm{i}\phi}$. Hence

$$F_2 = \sin\theta\,\mathrm{d}\theta \wedge \mathrm{d}\phi, \tag{37}$$

The unitary connection on the ket line is $\nabla_{\text{ket}} = \mathrm{d} + \mathrm{i}A_2$, whereas its dual has $\nabla_{\text{pre}} = \mathrm{d} - \mathrm{i}A_2$. Therefore

$$\boxed{c_1(L_{\text{ket}}) = -\frac{1}{2\pi}\int_{\mathbb{CP}^1} F_2 = -2, \qquad c_1(L_{\text{pre}}) = \frac{1}{2\pi}\int_{\mathbb{CP}^1} F_2 = 2.} \tag{38}$$

Equivalently, the transition function and Eq. (28) prove

$$\boxed{L_{\text{ket}} = \nu_2^*\mathcal{O}_{\mathbb{CP}^2}(-1) = \mathcal{O}_{\mathbb{CP}^1}(-2), \qquad L_{\text{pre}} = L_{\text{ket}}^* = \nu_2^*\mathcal{O}_{\mathbb{CP}^2}(1) = \mathcal{O}_{\mathbb{CP}^1}(2).} \tag{39}$$

Remark 5.1 (Why the dual is essential rather than notational). The transition $e^{-2\mathrm{i}\phi}$ belongs to the algebraic ket eigenline $\mathcal{O}(-2)$. Holomorphic geometric quantization uses its positive dual $\mathcal{O}(2)$. This is also the sign distinction between the microscopic ket kinetic term and the positive prequantum flux; it cannot be removed merely by renaming the same line bundle.

5.4 Three-state quantization

Holomorphic quantization of the dual prequantum line yields

$$\mathcal{H}_{|k|=2} = H^0(\mathbb{CP}^1, L_{\mathrm{pre}}) = H^0(\mathbb{CP}^1, \mathcal{O}(2)) = \mathrm{Span}\{z_0^2, z_0 z_1, z_1^2\} \simeq \mathrm{Sym}^2\,\mathbb{C}^2, \qquad \dim \mathcal{H}_{|k|=2} = 3. \tag{40}$$

The spin-one generators obey

$$[J_x, J_y] = \mathrm{i}J_z, \qquad J_x^2 + J_y^2 + J_z^2 = 2I_3. \tag{41}$$

The same symmetric square therefore appears in the exact spectrum, Berry line, and holomorphic state count.

6 Low-energy action and robustness

Adiabatic integration of gapped modes gives

$$S_{\mathrm{eff}}[s] = -2\hbar \int A_1(s) - \int H_{\mathrm{eff}}(s)\,\mathrm{d}t + O\left(\frac{\partial_t^2}{\Delta}\right), \tag{42}$$

where a conservative retained gap is

$$\Delta = \min\{\Delta_{\mathrm{ch}}, J_H, h, \Delta_{\mathrm{extra}}\}. \tag{43}$$

Indeed $\langle s|\mathrm{d}s\rangle = \mathrm{i}A_1$, so the microscopic kinetic term gives $\mathrm{i}\hbar\langle s;2|\mathrm{d}(s;2)\rangle = -2\hbar A_1$. Thus, relative to Eq. (2), the aligned $-h\boldsymbol{n}\cdot\boldsymbol{S}_{\mathrm{tot}}$ Hamiltonian has $k=-2$. Replacing it by the anti-aligned orientation (for example $+h\boldsymbol{n}\cdot\boldsymbol{S}_{\mathrm{tot}}$) reverses A_1 and gives $k=+2$. The magnitude two is topological and cannot vary under a smooth perturbation unless the line ceases to be isolated or its gap closes. Which sign is the physical Route-E chirality is not selected by the dimer itself.

At the benchmark, the verifier uses

$$(\Lambda_{\mathrm{portal}}, T, \hbar\omega) = (0.01, 0.02, 0.03), \tag{44}$$

while the smallest retained gap is $h = 0.2$. The ratio $0.03/0.2 = 0.15$ passes the declared 0.2 guard, whereas a portal scale 0.22 is rejected. This is a benchmark separation check, not a universal error bound.

7 Why the candidate still does not derive exactly two

There are two distinct meanings of a level-two-magnitude origin.

1. *Infrared origin:* after declaring two orbitals and working below all charge and singlet gaps, the retained ket/prequantum pair has Chern numbers $(-2, +2)$, hence $|k| = 2$. This note proves that statement.

2. *Fundamental selection:* the full UV theory explains why the number of same-orientation constituents is exactly two and prohibits every singleton or odd sector. This note does not prove that statement.

A single occupied orbital has ket/prequantum lines $\mathcal{O}(-1)$ and $\mathcal{O}(1)$, respectively, and a two-state doublet. The Mott model makes charge-removal states heavy, so they are absent from the strict IR EFT, but they remain states in the full microscopic theory. To claim that 2 is the fundamental lowest nonzero level, one needs at least one additional mechanism:

1. an exact microscopic gauge constraint whose physical local operators are necessarily bilinears and whose odd sectors are absent;
2. a discrete gauge superselection with an independently derived even sector and a plateau selecting its first nonzero representation;
3. an anomaly or generalized-symmetry condition fixing a coefficient to two in a specified four-dimensional UV completion;
4. a lattice or unit-cell theorem deriving exactly two locked orbitals and forbidding isolated-orbital boundary states.

This is the principal remaining AP-E3 blocker.

8 Alternative microscopic branches

8.1 Filled-fermion determinant

For r gapped fermion flavors, consider

$$H_f(\boldsymbol{n}) = \sum_{a=1}^{r} \left[-\Delta_a \psi_a^\dagger (\boldsymbol{n} \cdot \boldsymbol{\sigma}) \psi_a + U_a (N_a - 1)^2 \right]. \tag{45}$$

If every flavor is singly occupied and remains gapped, the many-body ground state is a Slater determinant of eigenlines. Berry connections add:

$$A_{\text{det}} = \sum_a A_a, \qquad k_{\text{ind}} = \sum_a C_a, \qquad C_a = \pm 1. \tag{46}$$

Two same-orientation filled lines give $|k| = 2$. One line gives $|k| = 1$, and opposite orientations cancel to zero. Flavor number, filling, and signs must all be derived. Recent experiments show that higher Chern sectors can be engineered and tuned, supporting possibility rather than necessity [6].

8.2 Mixed WZW matching

Let ω_3 and ω_2 be integrally normalized generators on the two target factors. A mixed Wess–Zumino–Witten term has schematic form

$$S_{\text{WZW}} = 2\pi\hbar n \int_{Y_5} \omega_3(g) \wedge \omega_2(s), \qquad n \in \mathbb{Z}. \tag{47}$$

For a localized configuration with baryon or skyrmion number B, reduction to its worldline gives

$$S_{\text{wl}} = \hbar n B \int_\gamma A, \qquad k = nB. \tag{48}$$

Davighi and Lohitsiri construct a weakly coupled UV completion in which the coefficient equals a color number and is tree-level exact [7]. Formally, $n_c = 2$ and $B = 1$ would give a signed $k = 2$. However, two-color theory has pseudoreal representations and a different Pauli–Guersey global symmetry and coset [8]. One cannot obtain a valid Route-E completion merely by replacing n_c with two; the global quotient, anomaly, minimal soliton, and mixed-WZW cohomology must be redone.

8.3 Tangent and Dirac route

The holomorphic tangent bundle satisfies

$$T^{1,0}\mathbb{CP}^1 \simeq \mathcal{O}(2). \tag{49}$$

If AP-E4 derives that a physical chiral fluctuation is tangent-valued, the canonical Spin^c or Dolbeault operator

$$D_T = \sqrt{2}(\bar{\partial}_T + \bar{\partial}_T^\dagger) \tag{50}$$

has index, with $x = c_1(\mathcal{O}(1))$,

$$\operatorname{ind} D_T = \int_{\mathbb{CP}^1} \operatorname{ch}(\mathcal{O}(2)) \operatorname{Td}(T\mathbb{CP}^1) = \int_{\mathbb{CP}^1} (1+2x)(1+x) = 3. \tag{51}$$

This convention must be explicit. An ordinary spin Dirac operator twisted by $\mathcal{O}(p)$ instead has positive-chirality holomorphic sections $H^0(\mathcal{O}(p-1))$, of dimension p for $p \geq 1$; at $p = 2$ that count is two, not three. Projective-space monopole spectra clarify these distinctions but take the monopole level as input [9]. The tangent/Dirac branch belongs to AP-E4 and cannot be used backward to close AP-E3.

9 Failure modes and falsifiable gates

Failure	Consequence and required check
$\mu \notin (0, U)$	The bare onsite ground changes to $N = 0$ or $N \geq 2$; repeat the complete occupancy minimization.
Interacting plateau inequality fails	The bare condition $0 < \mu < U$ no longer proves $(1, 1)$; minimize Eq. (13) globally.
$J_H < 0$	The singlet is the unique ground state and carries $k = 0$.
No locking	The target is $\mathbb{CP}^1 \times \mathbb{CP}^1$, not its diagonal $\mathbb{CP}^1$.
Opposite Berry orientations	The two Chern numbers cancel, yielding $k = 0$.
Singleton allowed	A physical $\lvert k\rvert = 1$ doublet remains; the fundamental exactly-two claim fails.
Orientation not derived	The declared aligned action has $k = -2$; obtaining the desired $k = +2$ requires a physical orientation/chirality argument.
$h = 0$	The triplet ground space is degenerate, so there is no unique Abelian ground-state line for the displayed Berry connection.
Gap-scale collision	If portal, temperature, or drive scales reach a charge, singlet, orientation, or extra-state gap, the one-line adiabatic EFT is not controlled.
Internal triplet only	A three-state spin multiplet is not automatically three chiral families; construct the four-dimensional Dirac operator and portal.

The strongest next AP-E3 tests are:

1. construct an explicit exactly-two gauge constraint or unit-cell symmetry and prove the absence of boundary singleton sectors;

2. embed Eq. (9) in an anomaly-consistent four-dimensional model;
3. compute the many-body Chern number with a lattice-link or Wilson-loop algorithm under every portal deformation up to the first gap closing [10];
4. prove that the Route-E coupling maps the triplet to chiral family data and does not introduce opposite-chirality partners.

10 The 27-check reproducible audit

The executable verifier is

`route_f/code/verify_ap_e3_level_two_microscopic.py`.

From the repository root, run

`python3 route_f/code/verify_ap_e3_level_two_microscopic.py`.

It writes JSON and Markdown reports in `route_f/output/`. The seed is 20260714. Its checks are grouped as follows.

Group	Count	Status	Content
M0 provenance	2	pass	Input hashes and green, non-promoting AP-E2 gate.
M1 Mott	4	pass	Bare window, false-inference counterexample, proved interacting plateau inequality, full minimum, and coercive tail.
M2 Hund	7	pass	Hermiticity, $SU(2)$, triplet/singlet spectrum, projector, gaps, sign reversal, and 64 orientations.
M3 Veronese	6	pass	Normalization, phase weight, Berry and metric factor two, ket/dual transition signs, and prequantum Chern quadrature.
M4 representation	2	pass	Spin-one Casimir and large-gauge non-selection.
M5 failures	3	pass	Opposite orientation, singleton, and unlocked target.
M6 EFT	2	pass	Positive hierarchy and rejected bad portal.
M7 boundary	1	pass	Magnitude proof complete; signed orientation and UV physics closure false.
Total	27	pass	

For $J_H = 1$, exact diagonalization gives

$$\operatorname{spec} H_{\text{Hund}} = (-0.25, -0.25, -0.25, 0.75), \tag{52}$$

while the antiferromagnetic sign gives $(-0.75, 0.25, 0.25, 0.25)$. Over 64 seeded orientations at $h = 0.2$, the maximum spectral residual is 8.88×10^{-16}. Over 256 random normalized sections, the maximum Veronese or norm residual is 9.99×10^{-16}, the Berry factor-two residual is 2.24×10^{-16}, and the metric residual is 2.22×10^{-16}.

For N_θ midpoint cells, direct curvature quadrature gives

$$c_1^{(N_\theta)} = \frac{\pi}{N_\theta} \sum_{j=0}^{N_\theta - 1} \sin\left(\frac{(j+1/2)\pi}{N_\theta}\right). \tag{53}$$

The deterministic sequence is

N_θ	$c_1^{(N_\theta)}$
20	2.0020576483
80	2.0001285163
320	2.0000080319
1280	2.0000005020

Its monotone second-order convergence to the dual-prequantum value two verifies Eq. (38) numerically; it does not replace the patch proof.

11 Recent-literature boundary

Recent primary work supplies useful ingredients but not the missing theorem:

- projective monopole spectra and twisted Dolbeault complexes support the AP-E4 spectral audit, while treating the level as an input [9];
- tunable higher Chern numbers demonstrate realizability but also show that a Hamiltonian can select different integers in different regimes [6];
- mixed-WZW generalized-symmetry matching can protect an integer coefficient, but its global symmetry and color content must match the UV completion [7];
- coupled-dimer coherent-state methods make singlet/triplet structure calculable, but antiferromagnetic pairing itself does not imply the ferromagnetic diagonal line or $|k| = 2$ [3].

No cited paper proves the repository-specific chain

$$\begin{gathered} \text{exactly two constituents} \Longrightarrow \text{no singleton}, \\ \Longrightarrow \{\mathcal{O}(-2)_{\text{ket}}, \mathcal{O}(2)_{\text{pre}}\} \Longrightarrow \text{three chiral Route-E families.} \end{gathered} \tag{54}$$

12 Conclusion and next decisions

Within the declared two-orbital Mott–Hund model, AP-E3 now has a complete first-principles magnitude chain:

$$\begin{gathered} C = \mu + h/2 + J_H/4 < U - J_H/8 \implies (N_1, N_2) = (1, 1), \\ J_H > 0 \implies \mathrm{Sym}^2\,\mathbb{C}^2 \text{ triplet}, \\ |s\rangle \otimes |s\rangle \implies A_2 = 2A_1, \\ L_{\text{ket}} = \nu_2^* \mathcal{O}(-1) = \mathcal{O}(-2), \qquad L_{\text{pre}} = L_{\text{ket}}^* = \mathcal{O}(2), \\ S_{\text{aligned}} = -2\hbar \int A_1, \qquad S_{\text{reversed}} = +2\hbar \int A_1, \\ |k| = 2 \implies \quad \dim H^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3. \end{gathered} \tag{55}$$

The analytic occupancy theorem and 27/27 numerical audit make this more than a formal analogy. The construction also has sharp counterexamples: antiferromagnetic exchange gives a $k = 0$ singlet, opposite orientations cancel, no locking leaves a product target, and a singleton restores $|k| = 1$. The aligned sign $k = -2$ also makes the Route-E chirality choice an explicit unresolved input rather than a hidden convention.

The candidate-level task is therefore complete. The full AP-E3 physics task is not. Three viable research choices remain:

1. **Preferred minimal route:** derive a microscopic gauge or unit-cell rule that makes the two-orbital composite fundamental and removes odd sectors, then construct the Route-E portal.

2. **Topological UV route:** build the correct $n_c = 2$ mixed-WZW completion and redo its global-symmetry and anomaly analysis.

3. **Independent AP-E4 route:** derive a tangent-valued chiral Dirac mode and its complete partner spectrum, without using the desired count as an input.

Until one of these closes its own gates,

$$\boxed{\begin{gathered}\texttt{candidate\_level\_magnitude\_derivation\_complete=true},\\ \texttt{signed\_route\_e\_level\_selected=false},\\ \texttt{ap\_e3\_physics\_closed=false},\\ \texttt{physics\_promotion\_allowed=false}.\end{gathered}} \tag{56}$$

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